

HICA-S, Track T4: Local Pruning of Bid-Independent Route Pools

Main results with complete proofs (reader-friendly version)

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Draft for supervisor review

October 2026

How to read this note

The procurement mechanism of HICA-S generates a route pool for every driver *before* any bid is observed, and then runs exact VCG on that pool. This note asks how far such a pool can be reduced, before bidding, by a rule that looks only at the driver whose routes are being deleted.

[Section 1](#) is self-contained. It lists the notation ([Table 1](#)), collects the elementary facts that are used repeatedly ([Section 1.2](#)), and then states and proves the three main results. In plain words:

- **Theorem 1 (local pruning frontier).** A route r of driver i can be deleted safely, using only driver i 's own data, exactly when for every admissible report a *local substitute* is at least as cheap: a route of the same driver that serves part of the orders of r , with the remaining orders sent to the fixed fleet (FD). Deleting all such routes changes no optimal value and no VCG payment, and every route that survives is needed in some instance, so no rule of this type can delete more.
- **Theorem 2 (price of locality).** The surviving pool can still contain one route for each of the $\Theta(n^B)$ bundles of at most B orders, even in instances in which none of these routes is ever used.
- **Theorem 3 (label rule).** Part of the test can be carried out while routes are being generated, so that partial routes are discarded early, without changing the surviving pool. (It reduces the number of label extensions, but in our implementation not the running time; see [Section 2.9](#).)

[Section 2.7](#) adds three results behind the final version of Algorithm A: the frontier can be computed inside the generation loop ([Proposition 3](#)); labels that can no longer be completed can be discarded without changing the pool ([Proposition 4](#)), which makes Algorithm A 1.9–3.9× faster on our instances; and a second label rule that compares generated labels is also safe ([Proposition 5](#)).

[Section 2](#) contains remarks, the procedures that compute these objects, the NP-hardness of exact non-local pruning, and the computational evidence. Nothing in [Section 1](#) depends on [Section 2](#). [Figure 1](#) shows how the results depend on each other.

1 Main results

1.1 Setting and notation

Orders and the fixed fleet. O is a finite set of orders. Every order $o \in O$ can always be delegated to the fixed-capacity fleet (FD) at a public price $q_o \geq 0$; for $A \subseteq O$ we write $q(A) := \sum_{o \in A} q_o$. The platform publishes a bundle cap $B \geq 1$.

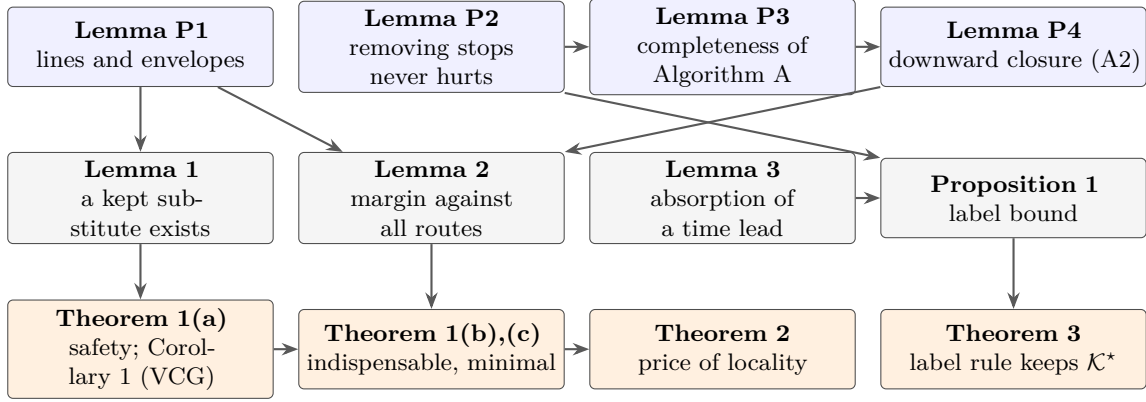


Figure 1: Roadmap. Blue boxes are preliminaries (Section 1.2), white boxes are the lemmas behind the main results, orange boxes are the main results. Theorem 3 also uses Lemma 1 and Lemma P3.

Drivers, routes and costs. N is the set of strategic drivers (gigworkers and occasional drivers). Driver i owns a finite route pool R_i , fixed before reports are collected. A route $r \in R_i$ serves a nonempty *bundle* $S_r \subseteq O$ with $|S_r| \leq B$ and has two public nonnegative coefficients, its *pair* $\text{pair}(r) = (K_r, W_r)$: K_r is a distance cost and W_r is working time in hours. If the driver reports the value of time $b \geq 0$, the reported cost of r is

$$c_r(b) = K_r + b W_r. \quad (1)$$

Thus every route is a *line* in the report b , with intercept K_r and slope W_r . The idle option of driver i is the empty route $\mathbf{0}_i$ with $S_{\mathbf{0}_i} = \emptyset$ and $K = W = 0$. We write $\text{pair}(r) \leq \text{pair}(r')$ if $K_r \leq K_{r'}$ and $W_r \leq W_{r'}$; then $c_r(b) \leq c_{r'}(b)$ for every $b \geq 0$.

Reports and the pruning domain. Pruning is required to be correct for all report profiles in a domain $\Theta = [\underline{\theta}, \bar{\theta}]$ with $0 \leq \underline{\theta} < \bar{\theta}$, announced before bidding (in the experiments $\Theta = [18, 25]$ money units per hour).

Winner determination and payments. An *allocation* $\rho = (\rho_j)_{j \in N}$ assigns to every driver one route $\rho_j \in R_j \cup \{\mathbf{0}_j\}$ such that the bundles are pairwise disjoint; orders that no selected route serves are sent to FD. For a report profile $b = (b_j)_{j \in N}$ its cost is

$$C(\rho; b) = \sum_{j \in N} c_{\rho_j}(b_j) + q\left(O \setminus \bigcup_{j \in N} S_{\rho_j}\right). \quad (2)$$

$V(R; b)$ is the minimum of $C(\rho; b)$ over all allocations; this is the set-partitioning winner-determination problem (Algorithm B), in which every order is served exactly once, by a route or by FD, and every driver receives at most one route. $V_{-k}(R; b)$ is the same minimum with driver k removed (Algorithm C). A winner k with route ρ_k^* is paid the Clarke-pivot payment

$$p_k = c_{\rho_k^*}(b_k) + V_{-k}(R; b) - V(R; b). \quad (3)$$

For sub-pools $R'_j \subseteq R_j$ we write $V(R'; b)$ for the same problem on the smaller pools.

Routes as stop sequences. Only the proofs of Lemmas P2 to P4 and of Theorems 2 and 3 look inside a route. There, a route of driver i is a sequence of stops (pickup p_o and delivery d_o of each

served order) starting from the point v_0 at the fixed departure time t_0 . At its k -th stop v_k the driver starts service and becomes ready to leave at

$$a_k = \max\{e_{v_k}, r_{k-1} + \tau(v_{k-1}, v_k)\}, \quad r_k = a_k + s_{v_k}, \quad r_0 = t_0, \quad (4)$$

where τ is travel time (minutes), s_v the service time and e_v the opening time of stop v (the release time for a pickup; $e_v = -\infty$ if the stop has no opening time). The max in (4) is the waiting time: a driver who arrives before e_v waits.

Standing assumptions. The following conditions hold throughout.

- (S1) Distances $d(\cdot, \cdot)$ and travel times $\tau(\cdot, \cdot)$ are nonnegative and satisfy the triangle inequality; service times are nonnegative.
- (S2) Opening and release times are lower bounds, met by waiting. Closing times, the end of availability, the capacity, the bundle cap B and (for an occasional driver) the detour budget are upper bounds.
- (S3) For a gigworker with distance coefficient κ , $K = \kappa \cdot (\text{route length})$ and $W = (\text{completion time} - t_0)/60$. For an occasional driver the route ends at her personal destination and $K = \kappa \cdot (\text{route length} - \text{direct length})$, $W = (\text{arrival time} - t_0 - \text{direct time})/60$; both are nonnegative by (S1).

Table 1 lists all symbols used in Section 1.

1.2 Preliminaries

Lines and lower envelopes. Because every cost (1) is a line in the report, the whole analysis reduces to comparing lines on an interval. The following facts are elementary and are used without further comment.

Lemma P1 (Lines and lower envelopes). *Let $g_1, \dots, g_m$ be lines in b and $E(b) = \min_j g_j(b)$ their lower envelope (Fig. 3a).*

- (i) *E is continuous, concave and piecewise linear. Its breakpoints are the values of b at which the lowest line changes.*
- (ii) *For any further line c , the difference $E - c$ is concave and piecewise linear with the same breakpoints. On an interval $[\underline{\theta}, \bar{\theta}]$ its maximum is attained at $\underline{\theta}$, at $\bar{\theta}$ or at a breakpoint in between.*
- (iii) *Two different lines agree at no more than one point. Hence a line that is nonnegative on an interval of positive length and is not identically zero is strictly positive on that interval except at, at most, one point.*
- (iv) *If finitely many strict inequalities between lines hold at b , they also hold at every b' close enough to b .*

Proof. A minimum of affine functions is concave; between two consecutive breakpoints $E - c$ is affine, so its maximum over each piece is at an end of the piece. The difference of two different lines is a nonzero affine function and has at most one root. Item (iv) is continuity. $\square$

Table 1: Notation.

<i>Data</i>	
$O; o, j, x$	set of orders; generic orders
$q_o, q(A)$	FD price of order o ; $q(A) = \sum_{o \in A} q_o$
B	bundle cap: a route serves at most B orders
N	set of strategic drivers (gigworkers and occasional drivers)
$\Theta = [\underline{\theta}, \bar{\theta}]$	pruning domain of reports, announced before bidding, $0 \leq \underline{\theta} < \bar{\theta}$
$b_i; b = (b_i)_{i \in N}$	report of driver i (value of time, per hour); report profile
<i>Routes and costs</i>	
R_i	route pool of driver i , fixed before reports
$\mathbf{0}_i$	idle route of driver i : no orders, $K = W = 0$
S_r	bundle of route r (set of orders it serves), $1 \leq S_r \leq B$
$\text{pair}(r) = (K_r, W_r)$	distance cost and working time (hours) of r
$c_r(b) = K_r + bW_r$	reported cost of r ; a line in b with intercept K_r , slope W_r
$\text{pair}(r) \leq \text{pair}(r')$	$K_r \leq K_{r'}$ and $W_r \leq W_{r'}$; implies $c_r(b) \leq c_{r'}(b)$ for all $b \geq 0$
<i>Allocation and payments</i>	
$\rho = (\rho_j)_{j \in N}$	allocation: one route or $\mathbf{0}_j$ per driver, disjoint bundles, the rest to FD
$C(\rho; b)$	cost (2) of allocation ρ at reports b
$V(R; b), V_{-k}(R; b)$	optimal value; optimal value without driver k
p_k	Clarke-pivot payment (3)
<i>Local frontier (Section 1.3)</i>	
$L_i = (O, q, B, \Theta, R_i)$	local view of driver i : everything a local rule may read
$D(r)$	local substitutes of r : other routes of the same driver serving a subset of S_r , and $\mathbf{0}_i$
$g_{r'}(b)$	substitute line $c_{r'}(b) + q(S_r \setminus S_{r'})$: use r' , send the rest of S_r to FD
$E_r(b)$	FD-completion envelope: the lowest substitute line at b
$\mu_r(b) = E_r(b) - c_r(b)$	local margin of r : how much r saves over its best local substitute
$\mathcal{K}_i^*$	local frontier of driver i : routes with $\mu_r(b) > 0$ for some $b \in \Theta$
Π^*	frontier rule: delete $R_i \setminus \mathcal{K}_i^*$ for every driver
<i>Routing (Lemmas P2–P4, Theorems 2–3)</i>	
p_o, d_o	pickup and delivery stop of order o
v_0, t_0	start point and (fixed) departure time of a driver
$d(u, v), \tau(u, v)$	distance (km) and travel time (min) from u to v
s_v, e_v, l_v	service time, opening time and closing time of stop v
a_k, r_k	service start and ready time at the k -th stop, (4)
κ	distance coefficient (money per km)
$L, P(L)$	label (a partial route) and the set of orders picked up in it
$\Delta D, \delta, A, \beta$	distance spent on T , time spent on T , absorption term, label bound (Definition 3)

Removing stops. The next lemma is the only place where routing enters the theory. Shortening a route obviously shortens its length; that it also never delays a later stop is less obvious, because a driver who arrives earlier may simply wait longer.

Lemma P2 (Removing stops never hurts). *Let σ' be obtained from a stop sequence σ by deleting some stops, both starting at (v_0, t_0) . Under (S1)–(S2):*

- (i) σ' is no longer than σ ;
- (ii) every stop kept in σ' starts service in σ' no later than in σ ;
- (iii) the completion time (for an occasional driver, the arrival time at the destination) of σ' is no later than that of σ .

Neither sequence needs to be feasible.

Proof. (i) is the triangle inequality, applied to every stretch of deleted stops. (ii) Let u and v be consecutive kept stops. In σ the driver travels from u to v through deleted stops, so, since waiting and service are nonnegative and by the triangle inequality, $a_\sigma(v) \geq a_\sigma(u) + s_u + \tau(u, v)$; moreover $a_\sigma(v) \geq e_v$. In σ' , by (4), $a_{\sigma'}(v) = \max\{e_v, a_{\sigma'}(u) + s_u + \tau(u, v)\}$. Hence $a_{\sigma'}(u) \leq a_\sigma(u)$ implies $a_{\sigma'}(v) \leq a_\sigma(v)$. The first kept stop is treated in the same way with $u = v_0$, and induction gives (ii). (iii) follows from (ii) applied to the last stop, or to the destination regarded as a final stop without opening time. $\square$

Algorithm A. Algorithm A generates the pool of each driver by forward label setting. A *label* is a partial route, stored as (v, IV, C, t, K, W) : current stop, orders on board, orders delivered, clock, and accumulated costs. Labels are extended one stop at a time and pass three filters (the final version, [Procedure 2](#) in [Section 2.7](#), processes the batches as tiers of equal event count and also computes the frontier inside the loop):

- *Layer 1 (label dominance).* Labels are processed in *batches*. Within a batch, a label is removed if another label with the same key (v, IV, C) has no later clock and no larger K and W , one of them strictly smaller. Every label that is created is filtered in exactly one batch, and its extensions are filtered in later batches.
- *Layer 2 (per-bundle Pareto).* Among completed routes with the same bundle, only the Pareto-minimal pairs (K, W) are kept, one route per pair.
- *Layer 3 (optional label rule).* The rule of [Section 1.6](#).

The returned pool R_i therefore contains no two routes with the same bundle and the same pair.

Lemma P3 (Completeness of Algorithm A). *Let ρ be any feasible route of driver i and R_i the pool returned by Algorithm A, possibly with Layer 3 switched on. Then*

- (a) R_i contains a route with bundle S_ρ and pair at most $\text{pair}(\rho)$; or
- (b) some feasible route $\tilde{\rho}$ with $S_{\tilde{\rho}} = S_\rho$ and $\text{pair}(\tilde{\rho}) \leq \text{pair}(\rho)$ passes through a label discarded by Layer 3.

Alternative (b) is impossible without Layer 3. In that case R_i contains exactly one route for each Pareto-minimal pair of each bundle.

Proof. Follow the prefixes of ρ (the labels it passes through) during the run, and look at the first prefix that does not survive.

Case 1: every prefix survives. Then ρ is completed, and Layer 2 keeps a route with bundle S_ρ and pair at most $\text{pair}(\rho)$: (a) holds.

Case 2: a prefix is discarded by Layer 3. Then (b) holds with $\tilde{\rho} = \rho$.

Case 3: a prefix L is removed by Layer 1. Dominance is irreflexive and transitive, so in the finite batch of L some label L' that dominates L survives. L' is at the same stop, has the same orders on board and the same orders delivered, and has no later clock and no larger costs. Append the remaining stops of ρ to L' . By (4), starting earlier never makes a later service start later, so every remaining stop is served no later than in ρ and all closing times are met. The added distance is the same and the completion (or arrival) time is no later, so the result is a feasible route ρ' with $S_{\rho'} = S_\rho$ and $\text{pair}(\rho') \leq \text{pair}(\rho)$. All prefixes of ρ' up to L' survive. Repeat the argument with ρ' in place of ρ : its first non-surviving prefix lies in a strictly later batch. As there are finitely many batches, Case 1 or Case 2 is eventually reached for some route whose pair is at most $\text{pair}(\rho)$, and (a) or (b) follows for ρ .

Without Layer 3, Case 2 never occurs, so (a) holds for every feasible route. A route kept by Layer 2 cannot be dominated by a feasible route ρ'' of the same bundle, because (a) for ρ'' would produce a kept route dominating it. Conversely, for a Pareto-minimal pair, (a) produces a kept route with exactly that pair. $\square$

Downward closure. The next property says that taking orders out of a route never makes it more expensive (Fig. 2). It is assumption A2 below, and it holds automatically for the pools of Algorithm A.

Lemma P4 (Pools of Algorithm A are downward closed). *Let R_i be produced by Algorithm A without Layer 3. For every $r' \in R_i$ and every $T \subseteq S_{r'}$ there is $\hat{r} \in R_i \cup \{\mathbf{0}_i\}$ with $S_{\hat{r}} = T$ and $\text{pair}(\hat{r}) \leq \text{pair}(r')$.*

Proof. For $T = \emptyset$ take $\hat{r} = \mathbf{0}_i$. Otherwise delete from r' the pickup and delivery of every order outside T , and call the result σ . By Lemma P2, σ is no longer than r' and serves every remaining stop no later; hence closing times, availability and (for an occasional driver) the detour budget are respected. The load at each remaining stop is that of the orders of T on board, which is at most the original load; precedence is inherited, and $|T| \leq B$. So σ is feasible, and $\text{pair}(\sigma) \leq \text{pair}(r')$ because both its length and its completion (or arrival) time are no larger. Lemma P3 without Layer 3 gives a route $\hat{r} \in R_i$ with bundle T and $\text{pair}(\hat{r}) \leq \text{pair}(\sigma) \leq \text{pair}(r')$. $\square$

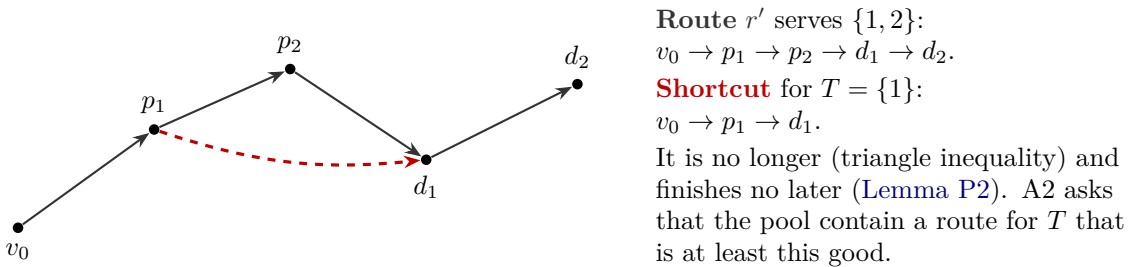


Figure 2: Downward closure (Assumption A2, Lemma P4): deleting the stops of some orders from a feasible route gives a feasible route for the remaining orders whose distance cost and working time are both no larger.

Truthfulness. The mechanism itself is exact VCG on a range that is fixed before bidding; its incentive properties are classical.

Lemma P5 (Truthfulness; Nisan and Ronen, 2007). *If the pools are fixed before reports are observed, every base and removal problem is solved exactly, ties are broken by a rule that reads no report, and costs have the form (1), then truthful reporting is a dominant strategy and every driver's utility is nonnegative.*

Proof. This is the maximal-in-range argument: the utility of driver i equals $V_{-i}(R; b)$ minus the cost of the selected allocation evaluated at i 's true cost and the others' reports, and the mechanism minimises exactly that cost when i reports truthfully. Utility is nonnegative because $\mathbf{0}_i$ is always available. See Nisan and Ronen (2007). $\square$

1.3 The local frontier

Definition 1 (Local view, local rule, safety). The *local view* of driver i is $L_i = (O, q, B, \Theta, R_i)$. A *local pruning rule* Π is a deterministic map that assigns to every local view a set $\Pi(L_i) \subseteq R_i$ of routes to delete; applied to every driver it yields the pools $R_j^\Pi = R_j \setminus \Pi(L_j)$. An *instance* is any finite set of drivers whose pools consist of routes with bundles of size at most B and nonnegative coefficients, in which no two routes of the same driver share both the bundle and the pair. The rule is *safe* if $V(R^\Pi; b) = V(R; b)$ for every instance and every $b \in \Theta^N$.

A local rule reads no report, so the pruned pools are fixed before bidding and Lemma P5 still applies. The ban on duplicates is harmless: duplicates can be merged before pruning without changing any optimal value, and Algorithm A never produces them.

Assumptions on the pool of the driver under study. Theorem 1(b)–(c) and Theorem 2 use the following three properties of a single pool R_i .

Assumption A1 (No duplicates). No two routes of R_i have the same bundle and the same pair.

Assumption A2 (Downward closure: dropping orders never costs more). For every route $r' \in R_i$ and every set $T \subseteq S_{r'}$ of its orders, $R_i \cup \{\mathbf{0}_i\}$ contains a route $\hat{r}$ with $S_{\hat{r}} = T$ and $\text{pair}(\hat{r}) \leq \text{pair}(r')$.

Assumption A3 (No free extra orders). If $r, r' \in R_i$ and $S_r \subsetneq S_{r'}$, then $\text{pair}(r) \neq \text{pair}(r')$: serving additional orders costs some distance or some time.

Assumption A1 holds in every instance by Definition 1; Assumptions A1 and A2 hold for every pool of Algorithm A by Lemmas P3 and P4 (see Fig. 2). Assumption A3 only excludes serving extra orders at exactly zero extra distance and zero extra time.

The envelope, the margin and the frontier. When the platform considers deleting a route r of driver i without looking at other drivers, the only replacements it can trust are those that involve nobody else: another route r' of driver i that serves *part* of S_r , with the remaining orders $S_r \setminus S_{r'}$ sent to FD at their public price. At report b such a replacement costs $g_{r'}(b) = c_{r'}(b) + q(S_r \setminus S_{r'})$, again a line in b .

Definition 2 (Substitutes, envelope, margin, frontier). For $r \in R_i$, the set of *local substitutes* is

$$D(r) = \{ r' \in R_i \cup \{\mathbf{0}_i\} : r' \neq r, S_{r'} \subseteq S_r \}.$$

The *FD-completion envelope* and the *local margin* of r are

$$E_r(b) = \min_{r' \in D(r)} \{ c_{r'}(b) + q(S_r \setminus S_{r'}) \}, \quad \mu_r(b) = E_r(b) - c_r(b).$$

The *local frontier* of driver i is $\mathcal{K}_i^* = \{ r \in R_i : \mu_r(b) > 0 \text{ for some } b \in \Theta \}$, and the *frontier rule* Π^* deletes $R_i \setminus \mathcal{K}_i^*$ for every driver.

In words, $\mu_r(b)$ is how much route r saves, at report b , compared with the best way of doing without r that uses only driver i and FD. A route belongs to the frontier if it is strictly the best local option for at least one admissible report. Since $\mathbf{0}_i \in D(r)$, always $E_r(b) \leq q(S_r)$. Figure 3 illustrates the definition. By Lemma P1(ii), membership in $\mathcal{K}_i^*$ is decided by evaluating μ_r at $\underline{\theta}$, at $\bar{\theta}$ and at the breakpoints of E_r in between (Procedure 1).

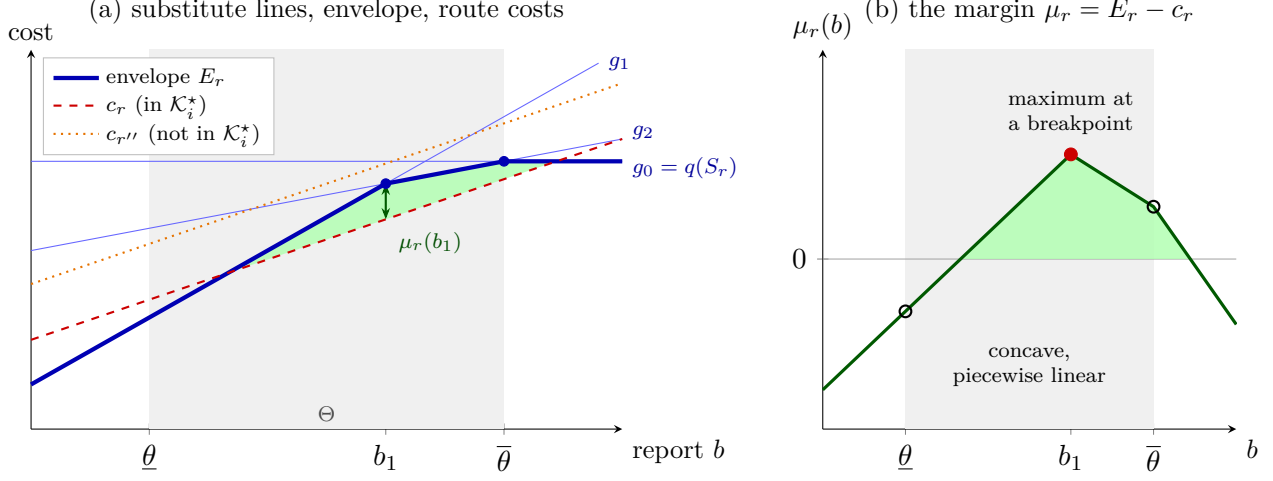


Figure 3: The envelope and the margin. (a) Each thin line is a local substitute $g_{r'}(b) = c_{r'}(b) + q(S_r \setminus S_{r'})$; g_0 is the idle option (all of S_r to FD). The thick line is their lower envelope E_r , with breakpoints (dots) where the lowest substitute changes. Route r (dashed) lies below E_r on the green region, so it is in the frontier. Route r'' (dotted) lies above E_r everywhere, so some local substitute is always at least as cheap and r'' can be deleted. (b) The margin $\mu_r = E_r - c_r$ is concave and piecewise linear, so its maximum over Θ is attained at $\underline{\theta}$, at $\bar{\theta}$ or at a breakpoint (Lemma P1); here at the breakpoint b_1 .

Example 1 (Twin instances). A gigworker g faces two orders with $q_{o_1} = 15$, $q_{o_2} = 45$, and $\Theta = [18, 25]$. Her pool has three routes:

$$r_a : \{o_1\}, (6, 0.6); \quad r_b : \{o_2\}, (8, 0.8); \quad r_c : \{o_1, o_2\}, (11, 1.2).$$

Frontier. r_a : its only substitute is $\mathbf{0}_g$, so $E_{r_a} = q_{o_1} = 15 < 6 + 0.6b$ on Θ ; $r_a \notin \mathcal{K}_g^*$ (FD is always cheaper). r_b : $E_{r_b} = 45 > 8 + 0.8b$ on Θ , so $r_b \in \mathcal{K}_g^*$. r_c : at $b = 18$ the three substitutes cost $6 + 10.8 + 45 = 61.8$ (r_a, o_2 to FD), $8 + 14.4 + 15 = 37.4$ (r_b, o_1 to FD) and 60 (idle), so $E_{r_c}(18) = 37.4 > 32.6 = c_{r_c}(18)$ and $r_c \in \mathcal{K}_g^*$.

Instance I_0 (g alone). r_c costs less than r_b plus FD for o_1 iff $11 + 1.2b < 23 + 0.8b$, i.e. $b < 30$; so r_c is optimal on all of Θ and r_b is never used.

Instance I_1 (I_0 plus a driver who serves o_1 at cost 0.5). At $b = 18$ the optimum is $22.4 + 0.5 = 22.9$ and uses r_b ; without r_b it is 32.6.

A rule that deletes r_b — for instance a hull across bundles, under which r_a dominates r_b — is right on I_0 and wrong on I_1 . Driver g 's data cannot tell the two instances apart. This is Theorem 1(b) in its smallest form.

1.4 Theorem 1: the local pruning frontier

Theorem 1 (Local pruning frontier). *Let $\underline{\theta} < \bar{\theta}$.*

(a) **Safety.** Π^* is safe. Moreover $V_{-k}(R^{\Pi^*}; b) = V_{-k}(R; b)$ for every driver k and every $b \in \Theta^N$.

- (b) **Indispensability.** Let R_i satisfy [Assumptions A2 and A3](#) and $r \in \mathcal{K}_i^*$. There are an instance that contains driver i with the same local view L_i , a report profile $b \in \Theta^N$ and a number $\gamma > 0$ such that every optimal allocation assigns r to i and $V(R \setminus \{r\}; b) \geq V(R; b) + \gamma/2$.
- (c) **Minimality.** Every safe local rule, applied to a local view that satisfies [Assumptions A2 and A3](#), deletes only routes outside $\mathcal{K}_i^*$. Hence $\mathcal{K}_i^*$ is the unique smallest pool that a safe local rule can retain, and Π^* attains it.

The proof of (a) rests on [Lemma 1](#): a deleted route can always be replaced by a route that is itself kept, not merely by some route that might have been deleted too. The proof of (b) rests on [Lemma 2](#), which extends the margin from substitutes (routes serving a subset of S_r) to *all* other routes of the driver.

Lemma 1 (A kept substitute always exists). *Let $\underline{\theta} < \bar{\theta}$. For every $r \in R_i \setminus \mathcal{K}_i^*$ and every $b \in \Theta$ there is $u \in \mathcal{K}_i^* \cup \{\mathbf{0}_i\}$ with $S_u \subseteq S_r$ and*

$$c_u(b) + q(S_r \setminus S_u) \leq c_r(b).$$

Proof. Fix b . Call every route $\rho \in R_i \cup \{\mathbf{0}_i\}$ with $S_\rho \subseteq S_r$ an *option* for S_r (this includes r itself), and let $\Phi(\rho) = c_\rho(b) + q(S_r \setminus S_\rho)$ be its cost at b : driver i serves S_ρ by ρ and FD serves the rest of S_r . Note $\Phi(r) = c_r(b)$.

Step 1: choose a cheapest option with a smallest bundle. Let m be the smallest value of Φ ; among the options with $\Phi = m$ take one with the fewest orders, and let T be its bundle and $k = |T|$. If $k = 0$, this option is $\mathbf{0}_i$ and $u = \mathbf{0}_i$ satisfies the claim, because $\Phi(\mathbf{0}_i) = m \leq \Phi(r) = c_r(b)$.

Step 2: a strict comparison at b . Let $k \geq 1$ and let M_T be the set of routes with bundle T and $\Phi = m$, i.e. with the same cost at b as the chosen option. We claim that every $\rho \in M_T$ is strictly cheaper at b than each of its own substitutes ρ'' that is not in M_T :

$$c_{\rho''}(b) + q(T \setminus S_{\rho''}) > c_\rho(b). \quad (5)$$

If $S_{\rho''} \subsetneq T$, then ρ'' is an option for S_r with fewer than k orders, so $\Phi(\rho'') > m = \Phi(\rho)$ by the choice of k ; removing the common term $q(S_r \setminus T)$ from both sides gives (5). If $S_{\rho''} = T$, then $\Phi(\rho'') \geq m$ means $c_{\rho''}(b) \geq c_\rho(b)$, and equality would put ρ'' in M_T .

Step 3: break ties by moving the report slightly ([Fig. 4](#)). The routes of M_T have the same bundle and the same cost at b . By [Assumption A1](#) their pairs differ, so their slopes W differ. If $b < \bar{\theta}$, let $u \in M_T$ be the route with the smallest slope. For $b' > b$ close to b , u is strictly cheaper than every other route of M_T , and the finitely many strict inequalities (5) still hold at b' ([Lemma P1\(iv\)](#)). So at b' route u is strictly cheaper than every one of its substitutes, that is $\mu_u(b') > 0$, and $u \in \mathcal{K}_i^*$. If $b = \bar{\theta}$, take the route with the largest slope and $b' < b$ instead.

Step 4: conclusion. $S_u = T \subseteq S_r$ and $c_u(b) + q(S_r \setminus T) = \Phi(u) = m \leq \Phi(r) = c_r(b)$. (Since $u \in \mathcal{K}_i^*$ and $r \notin \mathcal{K}_i^*$, automatically $u \neq r$.) $\square$

Proof of Theorem 1(a). Fix an instance, a profile $b \in \Theta^N$ and an optimal allocation ρ for the full pools. For every driver j whose route ρ_j is deleted ($\rho_j \notin \mathcal{K}_j^* \cup \{\mathbf{0}_j\}$), [Lemma 1](#) gives a kept route $u_j \in \mathcal{K}_j^* \cup \{\mathbf{0}_j\}$ with $S_{u_j} \subseteq S_{\rho_j}$ and $c_{u_j}(b_j) + q(S_{\rho_j} \setminus S_{u_j}) \leq c_{\rho_j}(b_j)$. Give u_j to driver j and send $S_{\rho_j} \setminus S_{u_j}$ to FD. Bundles only shrink, so they stay disjoint; every order is still served exactly once; and no other driver is affected, so all replacements can be made at the same time. The new allocation uses only kept routes and costs at most $V(R; b)$. Hence $V(R^{\Pi^*}; b) \leq V(R; b)$; the reverse inequality holds because the pruned pools are smaller. Removing a driver k gives another instance in which every remaining driver has the same local view, so the same argument applies to V_{-k} . $\square$

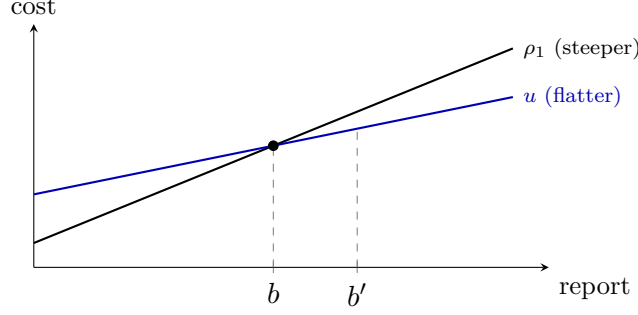


Figure 4: Tie-breaking in Step 3 of Lemma 1. Two routes with the same bundle cost the same at b . Just to the right of b the flatter one (the one with less working time) is strictly cheaper; just to the left, the steeper one. This is why ties cannot block membership in the frontier.

Corollary 1 (VCG outcomes are preserved). *Let $\mathcal{K}^* = \bigcup_{i \in N} \mathcal{K}_i^*$ and $b \in \Theta^N$.*

- (i) *Running Algorithms B and C on the pruned pools is VCG on a range fixed before bidding, hence truthful and individually rational (Lemma P5).*
- (ii) *Suppose the full-pool mechanism breaks ties among optimal allocations first by using as few routes outside $\mathcal{K}^*$ as possible, and then by the same report-independent rule as the pruned mechanism. Then both select the same allocation, and every payment (3) coincides.*
- (iii) *If the full problem has a unique optimal allocation at b , that allocation uses only routes of $\mathcal{K}^*$, and (ii) holds under any tie-breaking rule.*

Proof. (i) is Lemma P5. (ii) The proof of Theorem 1(a) turns every optimal allocation into one of no larger cost that uses only routes of $\mathcal{K}^*$. So the optimal allocations that use no route outside $\mathcal{K}^*$ are exactly the optimal allocations of the pruned problem, which has the same optimal value; the remaining tie-breaking criteria are the same in both mechanisms. V and every V_{-k} coincide by Theorem 1(a), so (3) coincides. The first tie-breaking criterion reads no report, so Lemma P5 also applies to the full-pool mechanism. (iii) If the unique optimum used a route outside $\mathcal{K}^*$, the replacement above would produce a different allocation of no larger cost. $\square$

Lemma 2 (Margin against all other routes). *Let R_i satisfy Assumptions A2 and A3, $\underline{\theta} < \bar{\theta}$, and $r \in \mathcal{K}_i^*$ with $S := S_r$. There are $b^* \in \Theta$ and $\gamma > 0$ such that*

$$c_{r'}(b^*) + q(S \setminus S_{r'}) \geq c_r(b^*) + \gamma \quad \text{for every } r' \in R_i \cup \{\mathbf{0}_i\}, r' \neq r. \quad (6)$$

In words: at report b^* , replacing r by any other route of driver i — whatever orders that route serves — and sending the orders of S that it misses to FD costs at least γ more. Orders outside S that r' happens to serve are not counted on the left; they are dealt with in the proof of Theorem 1(b).

Proof. Step 1: an interval of positive margin. Since $r \in \mathcal{K}_i^*$, $\mu_r(\bar{b}) > 0$ for some $\bar{b} \in \Theta$. The margin is continuous (Lemma P1), so $\mu_r > 0$ on a whole interval $J \subseteq \Theta$ of positive length around $\bar{b}$; if $\bar{b}$ is an end point of Θ , J lies on the inner side, which is possible because $\underline{\theta} < \bar{\theta}$.

Step 2: three cases for r' (Fig. 5). Let $b \in J$.

- (i) $S_{r'} \subseteq S$. Then r' is a substitute of r , so the left-hand side of (6) is at least $E_r(b) = c_r(b) + \mu_r(b)$.
- (ii) $S_{r'}$ overlaps S partially, i.e. $T := S \cap S_{r'} \neq S$ and $S_{r'} \not\subseteq S$. By Assumption A2 applied to r' and T , there is a route $\hat{r}$ with $S_{\hat{r}} = T$ and $c_{\hat{r}}(b) \leq c_{r'}(b)$. As $T \neq S$, $\hat{r}$ is a substitute of r . Since $S \setminus S_{r'} = S \setminus T$,

$$c_{r'}(b) + q(S \setminus S_{r'}) \geq c_{\hat{r}}(b) + q(S \setminus T) \geq E_r(b) = c_r(b) + \mu_r(b).$$

(iii) $S \subsetneq S_{r'}$. Then $q(S \setminus S_{r'}) = 0$. By [Assumption A2](#) with $T = S$ there is $\hat{r}$ with $S_{\hat{r}} = S$ and $c_{\hat{r}}(b) \leq c_{r'}(b)$. If $\hat{r} = r$, then $c_{r'}(b) \geq c_r(b)$. If $\hat{r} \neq r$, then $\hat{r}$ is a substitute of r with the same bundle, so $c_{\hat{r}}(b) \geq E_r(b) > c_r(b)$. In both cases the line $h_{r'}(b) := c_{r'}(b) - c_r(b)$ is nonnegative on J . By [Assumption A3](#), $\text{pair}(r') \neq \text{pair}(r)$, so $h_{r'}$ is not identically zero, and by [Lemma P1](#)(iii) it is strictly positive on J except at, at most, one point.

Step 3: choice of b^ and γ .* Pick $b^* \in J$ different from the finitely many exceptional points of case (iii); this is possible because J has positive length. Put $\gamma = \min\{\mu_r(b^*), \min_{r' \text{ in case (iii)}} h_{r'}(b^*)\} > 0$. Cases (i) and (ii) give (6) because $\mu_r(b^*) \geq \gamma$, and case (iii) because $h_{r'}(b^*) \geq \gamma$. $\square$

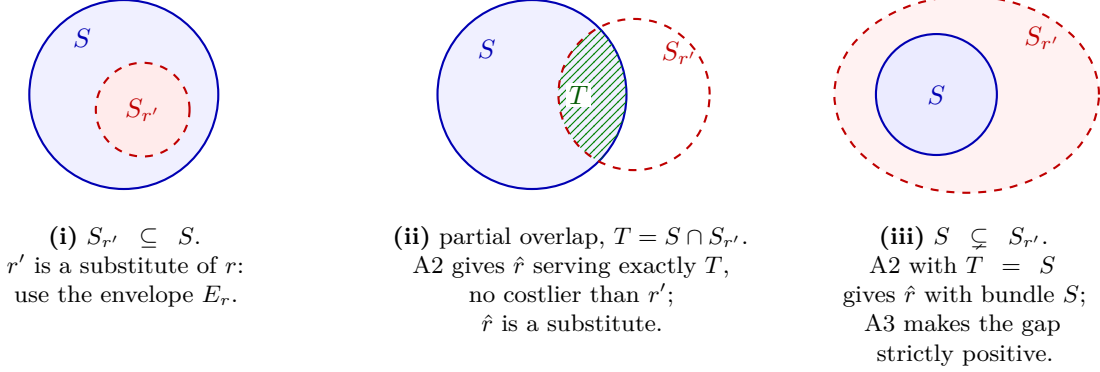


Figure 5: The three ways another route r' of the same driver can relate to the bundle $S = S_r$ (proof of [Lemma 2](#)). In (ii), downward closure replaces r' by a route $\hat{r}$ that serves only the common orders T and is no more expensive; $\hat{r}$ is then a substitute of r , so the envelope applies.

Proof of Theorem 1(b). Let b^* and γ be as in [Lemma 2](#) and fix $0 < \varepsilon \leq \gamma/(2B)$.

Construction (Fig. 6). The instance consists of driver i with report b^* and, for every order $x \notin S$, a helper driver j_x whose pool is a single route serving only $\{x\}$ with $K = \varepsilon$ and $W = 0$ (so its report does not matter). The data O, q, B, Θ, R_i are unchanged, so driver i has local view L_i .

Evaluation. Fix the route $r' \in R_i \cup \{0_i\}$ that driver i receives, and complete the allocation as cheaply as possible:

- orders of $S_{r'}$ are served by i ;
- orders of $S \setminus S_{r'}$ can only go to FD, since no helper serves them;
- every order $x \notin S \cup S_{r'}$ is served by its helper or by FD, at cost $e_x := \min\{\varepsilon, q_x\} \leq \varepsilon$;
- for orders $x \in S_{r'} \setminus S$ the helper stays idle.

The cheapest allocation in which i receives r' therefore costs

$$F(r') = c_{r'}(b^*) + q(S \setminus S_{r'}) + \sum_{x \notin S \cup S_{r'}} e_x, \quad \text{in particular} \quad F(r) = c_r(b^*) + \sum_{x \notin S} e_x.$$

Subtracting, and using (6) and $|S_{r'} \setminus S| \leq B$,

$$F(r') - F(r) = \underbrace{c_{r'}(b^*) + q(S \setminus S_{r'}) - c_r(b^*)}_{\geq \gamma} - \underbrace{\sum_{x \in S_{r'} \setminus S} e_x}_{\leq B\varepsilon \leq \gamma/2} \geq \frac{\gamma}{2} \quad \text{for every } r' \neq r.$$

Hence every optimal allocation gives r to driver i , and $V(R \setminus \{r\}; b) = \min_{r' \neq r} F(r') \geq F(r) + \gamma/2 = V(R; b) + \gamma/2$. $\square$

The helpers make every order outside S almost free. A route r' that serves such orders therefore gains almost nothing from them (at most $B\varepsilon$), while by Lemma 2 it loses at least γ on the orders of S .

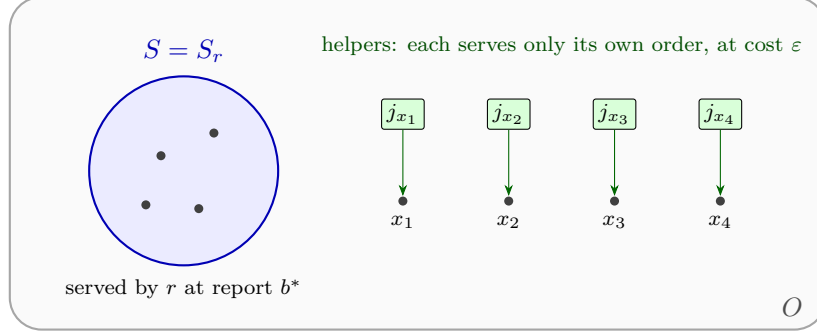


Figure 6: The instance in the proof of Theorem 1(b). Orders outside S become almost free (cost ε), orders inside S can only be served by driver i or by FD. Driver i 's own data are unchanged, so no local rule can distinguish this instance from the original one.

Proof of Theorem 1(c). Suppose a local rule Π deletes some $r \in \mathcal{K}_i^*$ from a local view L_i that satisfies Assumptions A2 and A3. In the instance of part (b), driver i has local view L_i , so Π deletes r there as well (and possibly other routes). Every allocation of the pruned pools is an allocation of $R \setminus \{r\}$, hence $V(R^\Pi; b) \geq V(R \setminus \{r\}; b) > V(R; b)$ and Π is not safe. So every safe local rule deletes only routes outside $\mathcal{K}_i^*$; Π^* deletes all of them and is safe by part (a). $\square$

1.5 Theorem 2: the price of locality

Theorem 1 says what a local rule must keep. Theorem 2 shows that this can be a great deal, and that all of it can be useless.

Theorem 2 (Price of locality). *Fix $B \geq 1$. For every $n \geq B$ there is a Euclidean instance I_n with n orders, one gigworker g and n occasional drivers such that*

- (i) R_g contains exactly one route for each bundle of at most B orders, all of them in $\mathcal{K}_g^*$; hence every safe local rule keeps $\sum_{k=1}^B \binom{n}{k} = \Theta(n^B)$ routes of g ;
- (ii) for every report profile in Θ^{n+1} , no optimal allocation assigns a route to g .

Construction (Fig. 7). All n orders are picked up at the same point P and delivered at the same point D , with $d(P, D) = \ell$, release time 0, non-binding deadlines, service time s (minutes) per stop and a common FD price q_0 . The gigworker starts at A , with $d(A, P) = a$, at time 0, with capacity at least B and distance coefficient $\kappa > 0$. Each occasional driver $\omega_1, \dots, \omega_n$ travels from P to D , departs at time 0 and has detour budget $2s$. Let $T_0 = \tau(A, P) + \tau(P, D)$. The parameters satisfy

$$q_0 > 2s\bar{\theta}/60, \tag{C1}$$

$$\kappa(a + \ell) + \underline{\theta}(T_0 + 2s)/60 < q_0, \tag{C2}$$

$$\kappa(a + \ell) > B(\bar{\theta} - \underline{\theta})2s/60. \tag{C3}$$

For example, $B = 3$, $s = 2$ min, $\Theta = [18, 25]$, $\kappa = 0.5$, $a = 2$ km, $\ell = 6$ km, speed 20 km/h and $q_0 = 20$ give $T_0 = 24$ min and (C1) $20 > 1.67$, (C2) $12.4 < 20$, (C3) $4 > 1.4$. (Condition (C3) depends on B ; with these numbers it holds for $B \leq 8$.)

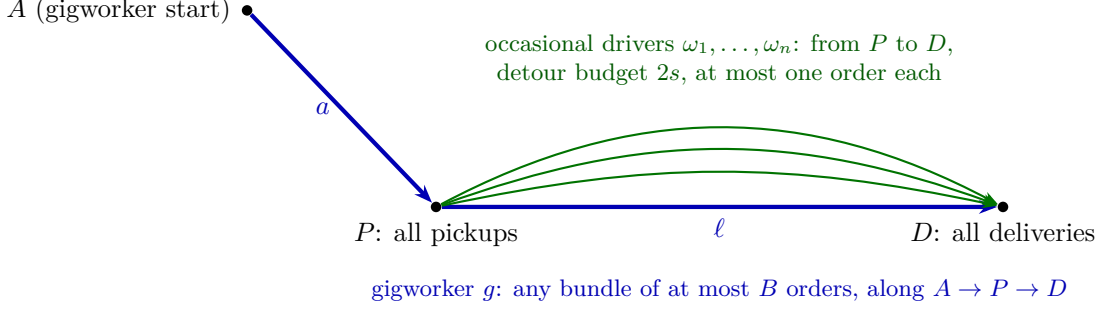


Figure 7: The instance of Theorem 2. The gigworker can serve every bundle of up to B orders along the same path, so her pool has one route per bundle, and none of these routes can be removed by a local rule. Yet the occasional drivers serve every order more cheaply, one each, so the gigworker never wins.

Proof. Step 1: the gigworker's pool. For a bundle S with $|S| = k \leq B$, every sequence that performs all pickups at P before all deliveries at D has length $a + \ell$ and completion time $T_0 + 2sk$ (no waiting). Every other sequence returns from D to P at least once, so it is at least 2ℓ longer and finishes later; it is Pareto-dominated. Hence $R_g = \{r_S\}$ with $K_{r_S} = \kappa(a + \ell)$ and $W_{r_S} = (T_0 + 2sk)/60$, one route per bundle. [Assumptions A1](#) and [A2](#) hold by [Lemma P4](#), and [Assumption A3](#) holds because W_{r_S} increases strictly with $|S|$.

Step 2: every route is in the frontier. Take $b = \underline{\theta}$ and a bundle S with $|S| = k$. The substitutes of r_S are $r_{S'}$ for $\emptyset \neq S' \subsetneq S$ and $\mathbf{0}_g$. For the former,

$$c_{r_{S'}}(b) + q_0(k - |S'|) - c_{r_S}(b) = (k - |S'|)(q_0 - 2sb/60) > 0$$

by (C1), since the two routes have the same K and differ by $2s(k - |S'|)$ minutes of working time. For the idle option,

$$q_0k - c_{r_S}(b) = (q_0 - c_{r_{\{x\}}}(b)) + (k - 1)(q_0 - 2sb/60) > 0$$

by (C2) and (C1), where $r_{\{x\}}$ is any single-order route. Hence $\mu_{r_S}(\underline{\theta}) > 0$, so $r_S \in \mathcal{K}_g^*$, and by Theorem 1(c) every safe local rule keeps it. This proves (i).

Step 3: the gigworker never wins. Each ω_j can serve any single order with zero extra distance and extra time $2s$ (the two service stops), and cannot serve two orders, which would need an extra time of at least $4s$. Take an allocation in which g serves r_S with $|S| = k$. The $n - k$ orders outside S occupy at most $n - k$ occasional drivers, so at least k of them are idle. Give the orders of S to k idle occasional drivers and make g idle. The reported cost changes by at most

$$-c_{r_S}(b_g) + k\bar{\theta} \frac{2s}{60} \leq -\kappa(a + \ell) - \underline{\theta} \frac{T_0 + 2sk}{60} + k\bar{\theta} \frac{2s}{60} \leq -\kappa(a + \ell) + B(\bar{\theta} - \underline{\theta}) \frac{2s}{60} < 0$$

by (C3). So no allocation that uses a route of g is optimal. This proves (ii). $\square$

The theorem makes the asymmetry between the driver classes precise: nothing bounds the pool that no local rule can shrink except feasibility. A tight detour budget restricts feasibility; time windows alone do not. [Remark 3](#) shows that an occasional driver with a generous detour budget is exposed in exactly the same way.

1.6 Theorem 3: an FD-completion label rule

The frontier is computed after Algorithm A has enumerated every route. This subsection moves part of the test inside the enumeration. When a partial route (a label) has picked up a set T of orders,

we bound from below how much any completion of it would save by dropping T and sending T to FD. If that saving is at least the FD price $q(T)$, the label and all its extensions are discarded. By Theorem 1(c) no rule of this kind can shrink the frontier; what is at stake is only the running time of Algorithm A.

Labels and shortcuts. Fix a driver with start v_0 , departure t_0 and coefficient κ . A label is identified with its stop sequence $L = (v_1, \dots, v_m)$, with service starts and ready times (4); D is its length and $r = r_m$ its ready time at its last stop v_m . $P(L)$ is the set of orders picked up in L .

Definition 3 (Shortcut quantities). Let $\emptyset \neq T \subseteq P(L)$, and let $L \setminus T$ be the subsequence of L without the stops of the orders of T , with last point v' , ready time r' there, and length D' ($v' = v_0$, $r' = t_0$, $D' = 0$ if $L \setminus T$ is empty). Let $\mathcal{F}(L)$ be the set of stops with an opening time that a feasible continuation of L may still visit: the pickups p_j of orders $j \notin P(L)$ with $r + \tau(v_m, p_j) \leq l_{p_j}$ (none if $|P(L)| = B$), and, if deliveries have opening times, the deliveries of orders on board at the end of L and of orders whose pickups are in $\mathcal{F}(L)$. Define

$$\begin{aligned} \Delta D(L, T) &= D - D' - d(v', v_m) && \text{(distance spent on } T), \\ \delta(L, T) &= r - r' - \tau(v', v_m) && \text{(time spent on } T), \\ A(L, T) &= \max\left\{0, \max_{w \in \mathcal{F}(L)} (e_w - r' - \tau(v', w))\right\} && \text{(time that later waiting can absorb)}, \\ \beta(L, T) &= \kappa \Delta D(L, T) + \underline{\theta} \frac{\max\{0, \delta(L, T) - A(L, T)\}}{60} && \text{(guaranteed saving)}. \end{aligned}$$

The *FD-completion label rule* discards L and all its extensions if $\beta(L, T) \geq q(T)$ for some $\emptyset \neq T \subseteq P(L)$.

The two corrections are what makes the rule safe (Fig. 8). The *junction terms* $d(v', v_m)$ and $\tau(v', v_m)$ account for the fact that the continuation of L starts at v_m , whereas the shortcut ends at v' : the shortcut still has to get to wherever the route goes next. The *absorption term* A accounts for waiting: a route that is ahead in time may simply wait longer at a later stop that has not opened yet, so part of its time saving can disappear (Fig. 9).

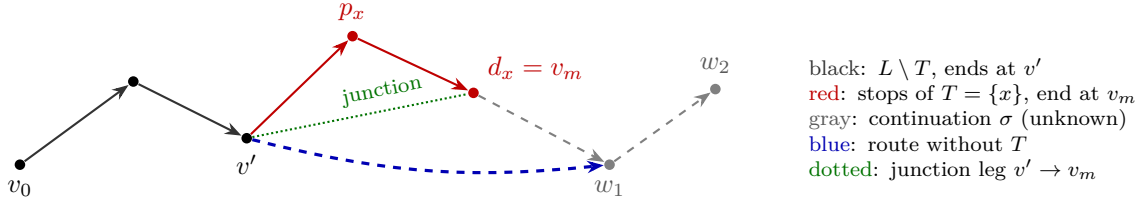


Figure 8: The shortcut of a label. Dropping order x replaces the red stretch and the first gray leg by the blue leg. The saving in distance is at least $D - D' - d(v', v_m)$, by the triangle inequality $d(v', w_1) \leq d(v', v_m) + d(v_m, w_1)$; this holds whatever the continuation is. Without the junction term the bound would credit the whole red stretch even when it lies on the way to w_1 .

Theorem 3 (The label rule preserves the frontier). *Let R_i be the pool of Algorithm A without Layer 3, and R'_i the pool obtained when Algorithm A in addition discards some or all labels that satisfy the rule, together with their extensions. Under the standing assumptions and $\underline{\theta} < \bar{\theta}$,*

$$\mathcal{K}_i^*(R'_i) = \mathcal{K}_i^*(R_i),$$

with routes identified by their bundle and pair. Consequently, applying the rule and then Π^ produces the same pool as Π^* alone, and Theorem 1 and Corollary 1 apply unchanged.*

The proof needs two ingredients: [Lemma 3](#), which quantifies how much of a time lead survives waiting, and [Proposition 1](#), which shows that β is a valid lower bound on the saving for every completion.

Lemma 3 (Absorption of a time lead). *Two paths visit the same stops $w_1, \dots, w_k$ in this order. Path 1 leaves point x at time η , path 2 leaves point x' at time η' . Let $T_1 = \eta' + \tau(x', w_1)$ and $T_{j+1} = T_j + s_{w_j} + \tau(w_j, w_{j+1})$ be the arrival times of path 2 if it never waited. If path 1 reaches w_1 at least δ later than T_1 , i.e. $\eta + \tau(x, w_1) \geq T_1 + \delta$, then for every j the service starts satisfy*

$$a_j^{(1)} - a_j^{(2)} \geq \delta - \max_{l \leq j} (e_{w_l} - T_l)^+.$$

In words: path 2 has a lead of δ . Waiting at a stop w_l can eat at most $(e_{w_l} - T_l)^+$ of that lead, namely the time by which path 2 would have arrived before the stop opens.

Proof. Write $\text{dur}(l \rightarrow j) = \sum_{h=l}^{j-1} (s_{w_h} + \tau(w_h, w_{h+1}))$ for the no-wait travel and service time from w_l to w_j . Unrolling (4) shows that a path arriving at w_1 at time U_1 starts service at w_j at

$$a_j = \max\{U_1 + \text{dur}(1 \rightarrow j), M_j\}, \quad M_j = \max_{l \leq j} \{e_{w_l} + \text{dur}(l \rightarrow j)\} :$$

either the driver never waited, or the start is forced by the last opening time at which she waited. M_j is the same for both paths. For path 2, $U_1 = T_1$ and $T_1 + \text{dur}(1 \rightarrow j) = T_j$, so $a_j^{(2)} = \max\{T_j, M_j\}$; for path 1, $U_1 \geq T_1 + \delta$, so $a_j^{(1)} \geq \max\{T_j + \delta, M_j\}$. If $M_j \leq T_j$ the difference is at least δ ; if $T_j < M_j \leq T_j + \delta$ it is at least $T_j + \delta - M_j$; otherwise it is at least 0. In every case it is at least $\delta - (M_j - T_j)^+$, and $M_j - T_j = \max_{l \leq j} (e_{w_l} - T_l)$ because $T_j = T_l + \text{dur}(l \rightarrow j)$. $\square$

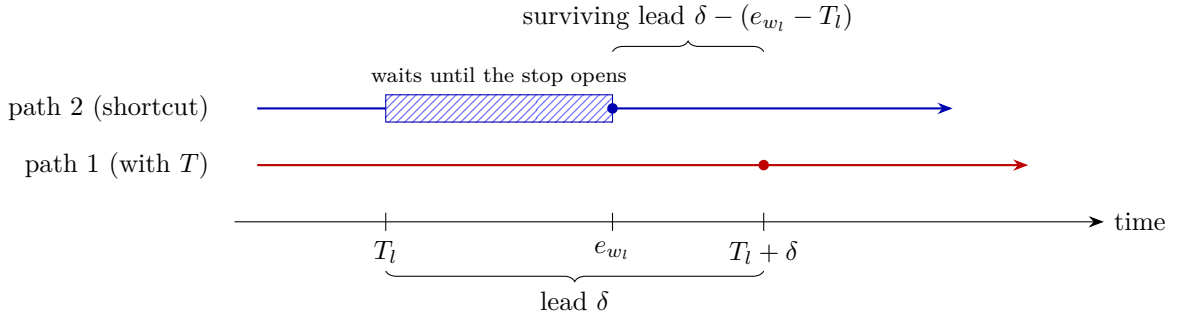


Figure 9: Absorption of a time lead ([Lemma 3](#)). The route without T (path 2) would reach stop w_l at T_l , but the stop opens only at e_{w_l} , so it waits. Of the original lead δ only $\delta - (e_{w_l} - T_l)$ survives. The term A in [Definition 3](#) bounds the largest such loss over all stops that a continuation may visit.

Proposition 1 (Label bound). *Let L be a label, $\emptyset \neq T \subseteq P(L)$, and ρ any feasible route of the driver that starts with L (possibly $\rho = L$). Let $\rho \setminus T$ be ρ without the stops of the orders of T . Then*

$$K_\rho - K_{\rho \setminus T} \geq \kappa \Delta D(L, T), \quad W_\rho - W_{\rho \setminus T} \geq \frac{\max\{0, \delta(L, T) - A(L, T)\}}{60},$$

and therefore $c_\rho(b) - c_{\rho \setminus T}(b) \geq \beta(L, T)$ for every $b \geq \underline{\theta}$. In particular, if the rule fires for (L, T) , then $c_\rho(b) \geq c_{\rho \setminus T}(b) + q(T)$ for every $b \in \Theta$: inside every completion of L , serving T costs at least as much as giving T to FD .

Proof. Write $\rho = L \cdot \sigma$, where σ is the continuation, and introduce the intermediate sequence $\pi = (L \setminus T) \cdot \sigma$ (the stops of T are removed from L only). Removing from π the stops of T that lie in σ (deliveries of orders of T still on board at the end of L) gives $\rho \setminus T$. By Lemma P2, $K_\pi \geq K_{\rho \setminus T}$ and $W_\pi \geq W_{\rho \setminus T}$, and also $W_\rho \geq W_{\rho \setminus T}$. So it suffices to show $K_\rho - K_\pi \geq \kappa \Delta D$ and $W_\rho - W_\pi \geq (\delta - A)/60$.

Distance. If σ is empty, the difference in length is $D - D' \geq \Delta D$. Otherwise let u be the first point of σ (for an occasional driver possibly the destination). The two sequences differ only up to u , so the difference in length is $D + d(v_m, u) - D' - d(v', u) \geq D - D' - d(v', v_m) = \Delta D$, because $d(v', u) \leq d(v', v_m) + d(v_m, u)$ (Fig. 8). For an occasional driver the same direct length is subtracted from both.

Time. If σ is empty, $W_\rho - W_\pi = (r - r')/60 \geq \delta/60$. Otherwise apply Lemma 3 to the stops $w_1, w_2, \dots$ of σ , with path 1 = ρ leaving v_m at r and path 2 = π leaving v' at r' . Its hypothesis holds: by the triangle inequality $\tau(v', w_1) \leq \tau(v', v_m) + \tau(v_m, w_1)$, so $r + \tau(v_m, w_1) \geq r' + \tau(v', w_1) + \delta$. At the last stop, or at the destination, the times therefore differ by at least $\delta - \max_l(e_{w_l} - T_l)^+$. It remains to show $\max_l(e_{w_l} - T_l)^+ \leq A$. A stop w_l of σ with an opening time is either a pickup of an order not picked up in L — which requires $|P(L)| < B$, and feasibility of ρ together with the triangle inequality gives $r + \tau(v_m, w_l) \leq l_{w_l}$ — or the delivery of an order on board or picked up later. In all cases $w_l \in \mathcal{F}(L)$. Since $T_l \geq r' + \tau(v', w_l)$ (triangle inequality, nonnegative service), we get $e_{w_l} - T_l \leq e_{w_l} - r' - \tau(v', w_l) \leq A$.

Cost. $c_\rho(b) - c_{\rho \setminus T}(b) = (K_\rho - K_{\rho \setminus T}) + b(W_\rho - W_{\rho \setminus T})$, and the second difference is nonnegative, so for $b \geq \underline{\theta}$ this is at least $\kappa \Delta D + \underline{\theta} \max\{0, \delta - A\}/60 = \beta$. $\square$

Proof of Theorem 3. Let $\mathcal{F}$ be the set of feasible routes of driver i . By Lemma P3, R_i contains exactly one route for each Pareto-minimal pair of each bundle, and every route of $\mathcal{F}$ satisfies alternative (a) or (b) of Lemma P3 with respect to R'_i . Write $D_R(r)$, E_r^R , μ_r^R for the substitutes, envelope and margin computed in a pool R .

Step 1: routes hit by the rule are outside the frontier. Let $\rho \in R_i$ satisfy alternative (b) with respect to R'_i : a feasible $\tilde{\rho}$ with the same bundle and $\text{pair}(\tilde{\rho}) \leq \text{pair}(\rho)$ passes through a label discarded with witness T . By Proposition 1, $c_{\tilde{\rho}}(b) \geq c_{\tilde{\rho} \setminus T}(b) + q(T)$ for all $b \in \Theta$. The route $\tilde{\rho} \setminus T$ is feasible (as in the proof of Lemma P4), so by Lemma P3 without Layer 3 there is $\hat{\rho} \in R_i \cup \{\mathbf{0}_i\}$ with bundle $S_{\tilde{\rho} \setminus T}$ and $c_{\hat{\rho}} \leq c_{\tilde{\rho} \setminus T}$. As $\hat{\rho}$ is a substitute of ρ ,

$$E_{\hat{\rho}}^{R_i}(b) \leq c_{\hat{\rho}}(b) + q(T) \leq c_{\tilde{\rho} \setminus T}(b) + q(T) \leq c_{\tilde{\rho}}(b) \leq c_{\rho}(b) \quad \text{for all } b \in \Theta,$$

so $\mu_{\hat{\rho}}^{R_i} \leq 0$ on Θ and $\rho \notin \mathcal{K}_i^*(R_i)$.

Step 2: every frontier route survives. Let $r \in \mathcal{K}_i^*(R_i)$. By Step 1, alternative (b) fails for r , so (a) holds: R'_i contains a route with bundle S_r and pair at most $\text{pair}(r)$. As r is Pareto-minimal, that pair is $\text{pair}(r)$, so $r \in R'_i$.

Step 3: frontier routes stay in the frontier. Let $r \in \mathcal{K}_i^*(R_i)$; then $r \in R'_i$ by Step 2. A substitute of r in R'_i is either a substitute of r in R_i , or a route $r_2 \notin R_i$. In the second case $\text{pair}(r_2)$ is not Pareto-minimal, so r_2 is dominated by some $r_1 \in R_i$ with the same bundle; and $r_1 \neq r$, because $r \in R'_i$ would otherwise have removed r_2 in Layer 2. So r_1 is a substitute of r in R_i and $c_{r_2}(b) + q(S_r \setminus S_{r_2}) \geq c_{r_1}(b) + q(S_r \setminus S_{r_1}) \geq E_r^{R_i}(b)$. Hence $E_{r_2}^{R'_i} \geq E_r^{R_i}$ on Θ , so $\mu_{r_2}^{R'_i} \geq \mu_r^{R_i}$ and $r \in \mathcal{K}_i^*(R'_i)$.

Step 4: no other route enters the frontier. Let $r \in R'_i$ with $r \notin \mathcal{K}_i^*(R_i)$. If $r \in R_i$, Lemma 1 (in R_i) gives for each b a route $u \in \mathcal{K}_i^*(R_i) \cup \{\mathbf{0}_i\}$ with $S_u \subseteq S_r$ and $c_u(b) + q(S_r \setminus S_u) \leq c_r(b)$; by Step 2, $u \in R'_i \cup \{\mathbf{0}_i\}$, and $u \neq r$. So $\mu_r^{R'_i} \leq 0$ on Θ . If $r \notin R_i$, then r is dominated by some $r_1 \in R_i$ with the same bundle. The pair of r_1 does not occur in R'_i (otherwise Layer 2 would have removed r), so alternative (a) fails for r_1 ; hence (b) holds and $r_1 \notin \mathcal{K}_i^*(R_i)$ by Step 1. Lemma 1 applied to r_1 gives

$u \in \mathcal{K}_i^*(R_i) \cup \{0_i\} \subseteq R'_i \cup \{0_i\}$ with $c_u(b) + q(S_r \setminus S_u) \leq c_{r_1}(b) \leq c_r(b)$, and $u \neq r$ because $r \notin R_i$. In both cases $r \notin \mathcal{K}_i^*(R'_i)$.

Steps 3 and 4 give $\mathcal{K}_i^*(R'_i) = \mathcal{K}_i^*(R_i)$. $\square$

Theorem 3 also covers the interaction with label dominance. A label removed by Layer 1 in favour of another label is never extended, so if an extension of the dominating label is later discarded by the rule, routes through the dominated label are lost although no label on their own path satisfies the rule. Lemma P3(b) says exactly that every such route is weakly dominated by a route through a discarded label, and Step 1 places it outside the frontier.

2 Remarks, algorithms and computational evidence

This section is not needed for the proofs of Section 1. It collects remarks on the scope of the results, the procedures that compute them, and the evidence from the audits and experiments.

2.1 What the frontier is not

Remark 1 (Three ways to beat the frontier). The minimality in Theorem 1(c) is relative to local rules (Definition 1). The frontier can be beaten in three ways, each of which leaves that class. (i) *Non-local rules* that also read other drivers' public data remain bid-independent and keep truthfulness; in the instances of Theorem 2 such a rule could delete every route of the gigworker. Deciding exactly what a non-local rule may delete is, however, NP-hard (Proposition 2). (ii) *Instance-specific rules*, which only need to be correct on the instance at hand, can keep far less: on five experimental instances only 0.4–1.5% of generated routes are optimal for any of 1,000 sampled report profiles, whereas the frontier keeps 3.2–23.1% of the pool. (iii) *Restricting the competitors* to geometrically realisable drivers weakens the safety requirement; Remark 2 bounds the effect.

Proposition 2 (Exact non-local pruning is hard). *For $B \geq 3$ it is NP-hard to decide, given the pools of all drivers, the prices q and a route r , whether $V(R \setminus \{r\}; b) = V(R; b)$ for every $b \in \Theta^N$, even if every route lies in the local frontier of its driver and no cost depends on the report.*

Proof. Reduction from exact cover by 3-sets (Garey and Johnson, 1979): a universe U with $|U| = 3m$ and a family $\mathcal{S}$ of 3-subsets of U . The orders are $U \cup \{o^*\}$, all with $q_o = 1$. For each $F \in \mathcal{S}$ a driver owns one route serving F at cost 0; a driver h owns one route for each single order of $U \cup \{o^*\}$, each at cost 0; and a driver i owns one route r serving $\{o^*\}$ with $K_r = \frac{1}{2}$, $W_r = 0$. Every route is cheaper than the FD price of its bundle and its driver has no route with a smaller bundle, so every route lies in its driver's frontier.

If an exact cover exists, the sets cover U and h serves o^* : $V(R) = V(R \setminus \{r\}) = 0$. Otherwise a maximum packing of $\mathcal{S}$ leaves $u \geq 3$ elements uncovered, and any other packing leaves at least $u + 3$. Without r , h serves one of the $u + 1$ uncovered orders and the rest go to FD, so the optimal value is u . With r , h serves an uncovered element and r serves o^* , giving $u - 1 + \frac{1}{2} = u - \frac{1}{2} < u$. Hence r can be deleted safely if and only if an exact cover exists. $\square$

Remark 2 (Realisable competitors). The helpers j_x in the proof of Theorem 1(b) are abstract. They can be replaced by genuine occasional drivers whose origin and destination are the pickup and delivery of x and whose detour budget equals the total service time s_x at the two stops. Under a general-position condition (no other order has its pickup and delivery, in this order, on the segment from p_x to d_x) such a driver can serve x and nothing else, with zero extra distance and working time $s_x/60$, so at report $\underline{\theta}$ its cost is $\lambda_x = \underline{\theta}s_x/60$ (1.20 in the audit setting with $s_x = 4$ min; 3.00 with the five-minute service of the experiments). Repeating the proof of Theorem 1(b) with

$\min\{\lambda_x, q_x\}$ in place of $\min\{\varepsilon, q_x\}$ shows that every frontier route whose margin γ in [Lemma 2](#) exceeds $\sum_{x \in S_{r'} \setminus S} \min\{\lambda_x, q_x\}$ for all r' (for instance $\gamma > B\theta \max_x s_x/60$) stays indispensable against such competitors. In a synthetic audit, 556 of 563 frontier routes (98.8%) did. Whether a safe local rule could exploit this credit in general is open: a gigworker route with waiting slack can absorb an extra order at zero extra working time.

2.2 Loose feasibility, not the driver class

Remark 3 (Theorem 2 for occasional drivers). The proof of Theorem 2 uses only that every bundle is feasible, that one sequence per bundle is Pareto-minimal, and that K is constant and W affine in $|S|$. It therefore applies to an occasional driver with origin A , destination D and detour budget at least $\tau(A, P) + \tau(P, D) - \tau(A, D) + 2sB$ in place of g . Her routes have $K_{r_S} = \kappa(a + \ell - d(A, D))$ and $W_{r_S} = (T_0 - \tau(A, D) + 2s|S|)/60$, and the proof holds with $a + \ell$ and T_0 replaced by $a + \ell - d(A, D)$ and $T_0 - \tau(A, D) \geq 0$. With $P = (0, 0)$, $D = (6, 0)$, $A = (0, 4)$ (km), $\kappa = 1$, and B, s, Θ , speed and q_0 as in the example, (C1)–(C3) hold (the left-hand side of (C3) is 2.79 against 1.40, and (C2) reads $6.50 < 20$), and a detour budget of 20.4 min suffices, below the 30 min used in the experiments. What Theorem 2 separates is loose from tight feasibility, not one driver class from the other.

Remark 4 (Choice of the evaluation point). The proof of Theorem 2 evaluates the margin at θ . Any $b \in \Theta$ works if (C2) is stated at that b ; the manuscript draft of 2026-09-30 uses $\bar{\theta}$ (giving $15.67 < 20$). Using θ gives the weakest condition.

2.3 Monotonicity and the choice of Θ

Remark 5 (Monotonicity). E_r is nondecreasing in every FD price q_o , and a larger Θ can only add reports at which $\mu_r > 0$. For a fixed pool, the frontier therefore grows with the FD tariff and with Θ : local pruning is strong when FD is cheap and weak when it is expensive. This is what the experiments show ([Table 5](#)).

Remark 6 (Reports outside Θ). The mechanism does not restrict reports. Because the pruned pools are fixed before bidding, the pruned mechanism is VCG on a fixed range and [Lemma P5](#) applies at every report profile, including profiles outside Θ^N . The domain matters only for *outcome equivalence* with the full pool, which holds at the truthful profile whenever the types lie in Θ . Taking $\Theta = [0, \infty)$ extends the equivalence to all reports at the price of a larger frontier; [Definition 2](#) is written with “for some b ” so that it covers this case.

2.4 Computing the frontier

Remark 7 (Complexity and numerical tolerance). $|D(r)| \leq 2^B f$, where f bounds the number of routes per bundle. By [Lemma P1\(ii\)](#), membership in $\mathcal{K}_i^*$ is decided by evaluating μ_r at the end points of Θ and at the breakpoints of E_r , in $O(|D(r)| \log |D(r)|)$ time; the frontier of a driver costs $O(|R_i| 2^B f \log(2^B f))$. In floating-point arithmetic the test should err on the safe side: delete a route only if the largest margin is below zero by more than a small tolerance. The frontier is a post-processing step: it does not speed up enumeration, but it shrinks every problem solved by Algorithms B and C.

2.5 Relation to LP duality and to the literature

Remark 8 (LP view). In the linear relaxation of the winner-determination problem let π_o be the dual of order o and $\sigma_i \leq 0$ that of driver i ’s one-route constraint. The FD columns impose $\pi_o \leq q_o$. If

Procedure 1 FRONTIER(R_i, q, Θ): the rule Π^* for one driver

Input: pool R_i with bundles S_r and pairs (K_r, W_r) ; FD prices q ; $\Theta = [\underline{\theta}, \bar{\theta}]$

Output: local frontier $\mathcal{K}_i^* \subseteq R_i$

```

1:  $\mathcal{K}_i^* \leftarrow \emptyset$ 
2: for each  $r \in R_i$  do
3:    $D(r) \leftarrow \{r' \in R_i \cup \{\mathbf{0}_i\} \setminus \{r\} : S_{r'} \subseteq S_r\}$ 
4:   for each  $r' \in D(r)$  form the line  $g_{r'}(b) = K_{r'} + q(S_r \setminus S_{r'}) + b W_{r'}$ 
5:   compute the lower envelope  $E_r = \min_{r' \in D(r)} g_{r'}$  on  $\Theta$   $\triangleright O(|D(r)| \log |D(r)|)$ 
6:    $\mathcal{B} \leftarrow \{\underline{\theta}, \bar{\theta}\} \cup \{\text{breakpoints of } E_r \text{ in } \Theta\}$ 
7:   if  $\max_{b \in \mathcal{B}} (E_r(b) - K_r - b W_r) > 0$  then  $\triangleright \mu_r$  is concave
8:     add  $r$  to  $\mathcal{K}_i^*$ 
9:   end if
10: end for
11: return  $\mathcal{K}_i^*$ 

```

$S_{r'} \subseteq S_r$ and $c_r \geq c_{r'} + q(S_r \setminus S_{r'})$, then, since both routes belong to driver i , σ_i cancels and

$$\bar{c}_r - \bar{c}_{r'} = c_r - c_{r'} - \pi(S_r \setminus S_{r'}) \geq \sum_{o \in S_r \setminus S_{r'}} (q_o - \pi_o) \geq 0$$

for every dual vector with $\pi \leq q$. So Π^* deletes a route when, at each admissible report, a route of the same driver has no larger reduced cost over the dual region cut by the outside option. Equivalently, r is *redundant* in the sense of Sol (1994), the combination consisting of r' and the FD columns of $S_r \setminus S_{r'}$. The bound $\pi \leq q$ is all a single driver can know about the duals; the construction of Theorem 1(b) realises $\pi \approx q$ on S_r and $\pi \approx 0$ elsewhere. Restricting $D(r)$ to routes of the same bundle and taking $\Theta = [0, \infty)$ turns $\mathcal{K}_i^*$ into the extreme supported points of each bundle's (K, W) frontier (Ehrgott, 2005).

Remark 9 (Literature). Column dominance in set covering (Beasley, 1987), the subsumed-subsets test of Müller (1998), and dominance and redundancy in column generation (Vanderbeck, 1994; Sol, 1994), as summarised by Lübbecke and Desrosiers (2005), all compare columns at known costs or at a fixed dual vector. Three features of Theorem 1 have no counterpart there: costs depend on a report ranging over a domain fixed by the mechanism designer, and the substitute may change with the report; safety is required for the integer optimum and for every Clarke-pivot removal problem; and the converse (Theorem 1(b)–(c)) shows that the frontier is minimal over all safe local rules. The label rule extends rollback pruning (Lozano et al., 2016), built on the dominance of Feillet et al. (2004); Algorithm A's Layer 1 is the forward dominance of Dumas et al. (1991).

2.6 Why the correction terms of the label rule are needed

Rollback pruning (Lozano et al., 2016) compares cost and time separately, so its faithful transfer credits no time saving at all: it is the rule of Definition 3 with $A = +\infty$, which is safe by Proposition 1 but gives up the time saving. Setting $A \equiv 0$ instead credits the whole time saving at the rate $\underline{\theta}$, as if working time were an additive arc cost. That is unsafe, because a route that arrives earlier may simply wait longer.

Example 2 (Crediting the whole time saving is unsafe). On a synthetic instance (gigworker g_0 , $B = 3$) consider the label $L = (p_{o_2}, d_{o_2})$ and $T = \{o_2\}$ with $q_{o_2} = 17.13$. Without the absorption term the bound credits a time saving of 42.9 minutes and gives $17.72 \geq q_{o_2}$, so the label would be

discarded. In the feasible completion $(p_{o_2}, d_{o_2}, p_{o_0}, d_{o_0})$, however, the pickup of o_0 opens late: without o_2 the driver would arrive early at p_{o_0} and wait. The true time saving is only 7.8 minutes and the true cost saving at $\underline{\theta}$ is $7.24 < 17.13$, so serving o_2 within this route is cheaper than delegating it. With the correction A the label is kept.

Example 3 (Why the junction terms are needed). If the start, p_x, d_x and the next pickup w lie on a line in this order, the label $L = (p_x, d_x)$ lies on the way to w . Dropping x saves only service time, and indeed $\Delta D = D - 0 - d(v_0, d_x) = 0$. Without the junction term the bound would credit the whole length of L .

A negative result. The natural alternative is to decide before a pickup is visited, using a precomputed bound on the extra distance and time over all possible next stops. This rule is safe, but on the synthetic instances it never fired: the median distance credit was 0.5–0.75 against FD prices of 15–21, and the median time credit was zero because the detour of about five minutes was entirely absorbed by waiting for later-released pickups (median absorption 34–46 minutes). Before the pickup is visited the continuation is unknown, and the bound must hold for the cheapest one. The rule of [Definition 3](#) evaluates the same idea one stop later, when the detour is known.

2.7 The final version of Algorithm A

The production code of Algorithm A processes labels in rounds of equal $|IV| + |C|$ and, inside a round, in repeated closures over deliveries; Layer 1 compares only labels created in the same round and the same closure step. Two labels with the same key (v, IV, C) can therefore be created in different batches and never be compared, and the dominated one is expanded (7.9–22.1% of the extension work on the five label instances, `EXPERIMENT_REPORT.md`). The final version processes labels in *tiers* of equal event count $m = |IV| + 2|C|$: every extension adds exactly one event, two labels with the same key have the same m , so every pair of same-key labels meets in one Layer-1 batch. Each label is filtered in exactly one tier and its extensions in later tiers, so [Lemma P3](#) and Theorem 3 hold verbatim. Moreover, all routes with k orders are complete once tier $2k$ has been processed, which allows the frontier to be computed inside the loop.

Proposition 3 (The frontier can be computed inside the loop). *Let R_i satisfy [Assumption A1](#) and $\underline{\theta} < \bar{\theta}$, and let $R_i(S)$ be its routes with bundle S . Examine the bundles in order of nondecreasing size and accept a route $r \in R_i(S)$ if and only if $\mu'_r(b) > 0$ for some $b \in \Theta$, where μ'_r is the margin of [Definition 2](#) computed over the restricted set of substitutes*

$$D'(r) = \{\mathbf{0}_i\} \cup \bigcup_{S' \subsetneq S} \mathcal{K}'(S') \cup (R_i(S) \setminus \{r\}),$$

and $\mathcal{K}'(S')$ is the set of routes already accepted for the bundle S' . Then the accepted routes are exactly $\mathcal{K}_i^*$.

Proof. By induction on $|S|$; assume $\mathcal{K}'(S') = \mathcal{K}_i^* \cap R_i(S')$ for every bundle with fewer orders than S , and let $r \in R_i(S)$. Since $D'(r) \subseteq D(r)$, the restricted envelope is no lower than E_r , so $\mu'_r \geq \mu_r$; if $r \in \mathcal{K}_i^*$, then $\mu'_r(b) > 0$ for some b and r is accepted. If $r \notin \mathcal{K}_i^*$, fix $b \in \Theta$. [Lemma 1](#) gives $u \in \mathcal{K}_i^* \cup \{\mathbf{0}_i\}$, $u \neq r$, with $S_u \subseteq S$ and $c_u(b) + q(S \setminus S_u) \leq c_r(b)$. Either $u = \mathbf{0}_i$; or $S_u \subsetneq S$, and then $u \in \mathcal{K}_i^* \cap R_i(S_u) = \mathcal{K}'(S_u)$ by induction; or $S_u = S$ and $u \in R_i(S) \setminus \{r\}$. In every case $u \in D'(r)$, so $\mu'_r(b) \leq 0$. As b was arbitrary, r is rejected. $\square$

The proposition needs only [Assumption A1](#), so it also applies to a pool generated with the label rule: by Theorem 3 that pool has the same frontier, and the in-loop computation returns it. In

EXPERIMENT_REPORT.md the tiered version with the in-loop frontier is called C5; its in-loop frontier was validated by tests only, and [Proposition 3](#) is the missing proof.

Removing dead ends. The production code checks the detour budget of an occasional driver only when the home leg is added, and checks a delivery’s closing time only when that delivery is added. Labels that can no longer be completed are therefore extended to the end; on the five label instances, occasional drivers account for 50–62% of all extension attempts, while their frontier holds 6–7 routes.

Proposition 4 (Removing dead ends). *Call a label $L = (v, IV, C, t, K, W)$ dead if $t + \tau(v, d_j) > l_{d_j}$ for some $j \in IV$, or if the driver is an occasional driver with home h and home deadline l_h , and*

$$\max\left\{t + \tau(v, h), \max_{j \in IV}(\max\{t + \tau(v, d_j), e_{d_j}\} + s_{d_j} + \tau(d_j, h))\right\} > l_h.$$

Discarding every dead label when it is created leaves R_i , and hence $\mathcal{K}_i^$, unchanged.*

Proof. Call a label *useless* if it has no feasible completion. (i) A dead label is useless: any completion reaches d_j no earlier than $t + \tau(v, d_j)$ and, for an occasional driver, reaches home no earlier than the left-hand side above (triangle inequality, nonnegative waiting and service). (ii) Extensions of a useless label are useless, since their completions are completions of the label. (iii) A label dominated by a useless label is useless: if L' , with the same key as L and no earlier clock, had a feasible completion σ , then $L \cdot \sigma$ would be feasible by (4). By (ii), the filter removes only useless labels and prevents only useless labels from being created. By (iii), a label that is not useless can be dominated only by labels that are not useless; by induction over the batches, these are present with and without the filter, so exactly the same labels that are not useless survive Layer 1. Every completed route comes from such labels, so the completed routes, the Pareto fronts of Layer 2 and therefore R_i are unchanged. $\square$

Tiers, the dead-end filter and the in-loop frontier together give Algorithm A in its final form ([Procedure 2](#)). It never stores R_i and outputs $\mathcal{K}_i^*$ directly; [Section 2.9](#) reports its running time.

A second safe label rule. The label rule of [Section 1.6](#) compares a label with its own virtual shortcut. One can instead compare it with another label that has actually been generated.

Proposition 5 (FD-dominance between generated labels). *Let $L_1 = (v, IV, C_1, t_1, K_1, W_1)$ and $L_2 = (v, IV, C_2, t_2, K_2, W_2)$ be labels of the same driver at the same stop with the same orders on board, where K and W are the accumulated distance cost and time of Algorithm A. Suppose $C_1 \subsetneq C_2$ and $t_1 \leq t_2$, let $T = C_2 \setminus C_1$ and $A = \max\{0, \max_{w \in \mathcal{F}(L_2)}(e_w - t_1 - \tau(v, w))\}$ with $\mathcal{F}$ as in [Definition 3](#). If*

$$(K_2 - K_1) + \underline{\theta} \frac{\max\{0, t_2 - t_1 - A\}}{60} \geq q(T),$$

then discarding L_2 and all its extensions leaves $\mathcal{K}_i^$ unchanged.*

Proof. Let $\rho_2 = L_2 \cdot \sigma$ be a feasible route through L_2 and $\rho_1 = L_1 \cdot \sigma$. The continuation σ starts at v , delivers the orders of IV and serves only orders outside $C_2 \cup IV$. Started at the same stop no later, it serves every stop no later (by (4)), so all closing times and the home deadline are met; the load along σ is the same; and ρ_1 serves fewer orders. Hence ρ_1 is feasible and serves $S_{\rho_2} \setminus T$. The distance added by σ is the same, so $K_{\rho_2} - K_{\rho_1} = K_2 - K_1$ (for an occasional driver the same direct trip is subtracted from both). [Lemma 3](#), applied to two paths that leave the same point v at t_2 and t_1 (lead $\delta = t_2 - t_1$, no junction term), and the argument of [Proposition 1](#) for the stops with an opening time give $W_{\rho_2} - W_{\rho_1} \geq \max\{0, t_2 - t_1 - A\}/60$. Therefore $c_{\rho_2}(b) \geq c_{\rho_1}(b) + q(T)$ for every $b \geq \underline{\theta}$. This is the conclusion of [Proposition 1](#) with the feasible route ρ_1 in place of $\rho \setminus T$, and the proof of Theorem 3 goes through unchanged. $\square$

Procedure 2 GENERATEROUTES(i): Algorithm A for one driver (final version: event tiers, dead-end filter, frontier inside the loop)

Input: public data of driver i ; orders O ; cap B ; FD prices q ; $\Theta = [\underline{\theta}, \bar{\theta}]$; flag $useRule$

Output: local frontier $\mathcal{K}_i^*$

```

1:  $\mathcal{L} \leftarrow \{(v_0, \emptyset, \emptyset, t_0, 0, 0)\}$ ;  $m \leftarrow 0$ ;  $\mathcal{C}[S] \leftarrow \emptyset$  and  $\mathcal{K}'(S) \leftarrow \emptyset$  for every bundle  $S$ 
2: while  $\mathcal{L} \neq \emptyset$  do ▷  $\mathcal{L}$  = labels with  $m = |IV| + 2|C|$  events
3:    $\mathcal{L} \leftarrow \text{FILTERDOMINATED}(\mathcal{L})$  ▷ Layer 1: one batch per tier, key  $(v, IV, C)$ 
4:   if  $useRule$  then  $\mathcal{L} \leftarrow \{\ell \in \mathcal{L} : \text{not RULEFIRES}(\ell)\}$  ▷ Layer 3, optional (Procedure 3)
5:   end if
6:    $\mathcal{L}' \leftarrow \emptyset$ 
7:   for each label  $\ell \in \mathcal{L}$  do
8:     if  $IV(\ell) = \emptyset$  and  $C(\ell) \neq \emptyset$  then
9:       add  $\ell$  (gigworker) or TRYHOME( $\ell$ ) (occasional driver, if feasible) to  $\mathcal{C}[C(\ell)]$ 
10:    end if
11:    add every feasible TRYDELIVERY( $\ell, j$ ),  $j \in IV(\ell)$ , that is not dead to  $\mathcal{L}'$  ▷ Proposition 4
12:    if  $|IV(\ell)| + |C(\ell)| < B$  then
13:      add every feasible TRYPICKUP( $\ell, j$ ),  $j \notin IV(\ell) \cup C(\ell)$ , that is not dead to  $\mathcal{L}'$ 
14:    end if
15:  end for
16:  if  $m \geq 2$  is even then ▷ every route with  $k = m/2$  orders is now complete
17:    for each bundle  $S$  with  $|S| = m/2$  and  $\mathcal{C}[S] \neq \emptyset$  do
18:       $F \leftarrow \text{PARETOFRONT}(\{(K_\ell, W_\ell) : \ell \in \mathcal{C}[S]\})$  ▷ Layer 2
19:       $\mathcal{K}'(S) \leftarrow \{r \in F : \mu'_r(b) > 0 \text{ for some } b \in \Theta\}$  ▷ Proposition 3, test as in Procedure 1
20:    end for
21:  end if
22:   $\mathcal{L} \leftarrow \mathcal{L}'$ ;  $m \leftarrow m + 1$ 
23: end while
24: return  $\mathcal{K}_i^* = \bigcup_S \mathcal{K}'(S)$ 

```

Proposition 5 is Idea 3 (variants C6/C7) of `EXPERIMENT_REPORT.md`, where it was validated by a completion-level audit but not proved.

2.8 Implementation of the label rule

Each label carries, for every picked order j , the triple (D'_j, v'_j, r'_j) of its shortcut $L \setminus \{j\}$. Extending the label by a stop u updates each triple in constant time: if u is the pickup of j the triple is initialised with the parent label; if u is the delivery of j it is unchanged; otherwise the shortcut travels to u exactly like the label. The test first evaluates β with $A = 0$, which is an upper bound, and computes A only if that bound reaches q_j . The overhead per extension is $O(B)$ plus rare $O(n)$ evaluations of A . Subsets T with $|T| \geq 2$ added almost nothing in the experiments and are not used.

Procedure 3 RULEFIRES(ℓ'): label rule after extending ℓ by stop u (Layer 3)

Input: new label ℓ' with length D , last stop $v_m = u$, ready time r , picked set $P(\ell')$; parent's shortcut triples (D'_j, v'_j, r'_j)

Output: TRUE if ℓ' and its extensions can be discarded

```

1: for each  $j \in P(\ell')$  do                                      $\triangleright$  update shortcut triples,  $O(1)$  each
2:   if  $u = p_j$  then
3:      $(D'_j, v'_j, r'_j) \leftarrow$  length, last stop and ready time of the parent  $\ell$ 
4:   else if  $u$  is not a stop of  $j$  then
5:      $D'_j \leftarrow D'_j + d(v'_j, u); \quad r'_j \leftarrow \max\{e_u, r'_j + \tau(v'_j, u)\} + s_u; \quad v'_j \leftarrow u$ 
6:   end if                                                      $\triangleright$  if  $u = d_j$  the shortcut is unchanged
7: end for
8: for each  $j \in P(\ell')$  do                                      $\triangleright$  test  $T = \{j\}$ 
9:    $\Delta D \leftarrow D - D'_j - d(v'_j, v_m); \quad \delta \leftarrow r - r'_j - \tau(v'_j, v_m)$ 
10:  if  $\kappa \Delta D + \underline{\theta} \max\{0, \delta\}/60 < q_j$  then continue          $\triangleright$  cheap upper bound with  $A = 0$ 
11:  end if
12:   $A \leftarrow \max\{0, \max_{w \in \mathcal{F}(\ell')} (e_w - r'_j - \tau(v'_j, w))\}$ 
13:  if  $\kappa \Delta D + \underline{\theta} \max\{0, \delta - A\}/60 \geq q_j$  then return TRUE
14:  end if
15: end for
16: return FALSE

```

2.9 Computational evidence

Synthetic audit of the theory. The statements of [Section 1](#) were checked against a reference implementation that shares no code with the production implementation: it enumerates every precedence-feasible sequence of every bundle, computes the frontier analytically and solves every winner-determination problem exactly (HiGHS, zero gap). The instances have $n = 8$ orders in a 10×10 km square, two gigworkers and two occasional drivers, $B = 3$, $\Theta = [18, 25]$, FD tariff $8 + 3 \max\{0, d - 2\}$, $\kappa = 0.5$, speed 20 km/h and two minutes of service per stop; nine seeds. These checks test the mathematics; they are not thesis results.

In both mutations the optimal values on 40 random report profiles per seed were still identical: the wrongly deleted routes are rarely optimal, so sampling reports would not have revealed the errors. The completion-level audit and the frontier comparison did.

Experimental instances: safety and pool reduction. On 600 instances of the main grid (alignment 0.50 and 0.90, all sizes and supply mixes), every base and removal problem was solved on

Table 2: Synthetic audit of Theorems 1 and 2.

Check	Result
Violations of Assumption A2 (all subsets, three reports)	0
Routes after Pareto filtering / kept by Π^*	2,985 / 1,157 (38.8%)
gigworkers / occasional drivers	1,123 of 2,908 / 34 of 77
Theorem 1(a): max. $ V(R^{\Pi^*}) - V(R) $ over base and removal solves	5.7×10^{-14}
Corollary 1 : max. payment difference (1,737 payments)	5.7×10^{-14}
Theorem 1(b): frontier routes made indispensable by the construction	1,157 / 1,157
smallest increase of the optimal value when the route is removed	3.4×10^{-4}
Remark 2 : indispensable against geometric competitors (four seeds)	556 / 563
Frontier routes deleted by a per-driver hull across bundles	1,144 / 1,157
Routes optimal for some of 400 report profiles (three seeds)	1.9–3.8% of pool, all in $\mathcal{K}^*$
Theorem 2, $n = 3, 5, 7, 9$: $ \mathcal{K}_g^* = \sum_k \binom{n}{k}$ and g never optimal	confirmed
Mutation: FD term removed from E_r	detected
Independent grid-based margin (5,000 points) vs. breakpoint method	42 / 42 pools agree

Table 3: Synthetic audit of the label rule (Theorem 3).

Check	$B = 3$ (3 seeds)	$B = 4$ (2 seeds)
Rule firings audited	5,868	33,292
Feasible completions checked against Proposition 1	9,975	87,744
Bound violations	0	0
Frontier identical with and without the rule	yes	yes
Max. optimal-value difference (random reports)	2.8×10^{-14}	1.3×10^{-13}
Mutation $A \equiv 0$	1,339 violations, frontier changed	–
Mutation without junction terms	4,173 violations, frontier changed	–

the full pool and on the frontier. Across 1,924 solves per variant, the largest deviation in any optimal value or payment was 1.1×10^{-13} . Table 4 reports the effect on the payment step.

Table 4: Payment computation on the full pool (naive) and on the frontier (600 instances, $B = 3$). Times in seconds, medians.

Alignment	n	Pool removed	t_C naive	t_C frontier	Speed-up [IQR]	Incl. construction
0.50	10	100.0%	0.039	0.003	11.20 [10.11, 13.14]	1.13
0.50	15	100.0%	0.086	0.005	17.98 [14.62, 22.05]	0.63
0.50	20	100.0%	0.186	0.006	30.40 [21.92, 41.85]	0.60
0.90	10	75.2%	0.134	0.060	2.34 [1.69, 3.04]	1.44
0.90	15	76.6%	0.248	0.144	1.81 [1.56, 2.57]	1.05
0.90	20	77.2%	0.655	0.259	2.72 [2.11, 3.37]	1.24
Pooled over 600 instances					3.01	1.06

The frontier as a pre-auction certificate. An empty frontier certifies, before any bid is collected, that a driver has no route worth considering at any admissible report. Table 5 separates this from the trivial case of a driver without any feasible route. Consistently with Remark 5, at alignment 0.50 and $n = 15$ the certified share among drivers with a feasible route falls from 92.3% to 59.2% and 23.1% as the FD tariff multiplier rises from 0.75 to 1.00 and 1.25, while the share without a feasible route stays at 43.7%.

Table 5: Drivers without a feasible route, and drivers certified in advance as dominated by FD ($R_i \neq \emptyset$, $\mathcal{K}_i^* = \emptyset$), main grid (7,500 drivers).

Alignment	0.10	0.30	0.50	0.70	0.90
No feasible route (% of drivers)	14.8	12.3	43.1	2.5	0.1
Certified FD-dominated (% of drivers)	20.4	19.5	34.3	10.5	1.0
Certified, among drivers with a feasible route (%)	23.9	22.3	60.2	10.7	1.0
Instances in which no auction is needed (%)	8.3	7.7	53.0	1.7	0.0

The label rule: safety. The rule was evaluated on five instances with $n \in \{10, 12, 15\}$ at $B = 3$ and $B = 4$. Every firing was checked against every feasible completion of the discarded label: 114,627 completions at $B = 3$ and 5,539,924 at $B = 4$, with no violation of Proposition 1 and no change of the frontier for any of the 15 drivers. Removing the absorption term produced 4,279 violations at $B = 3$ and 123,237 at $B = 4$; removing the junction terms produced 3,149 and 96,112; both mutations changed the frontier of two to four gigworkers. The audit was repeated inside the tiered version of Algorithm A (the rule added to C5): 11,238 firings, 8,589 completions, no violation; both mutations were detected again (EXPERIMENT_REPORT_C8.md). On the synthetic instances the rule keeps about half of the labels that lie on a path to some feasible route, whereas an ideal local rule would keep 2.7% ($B = 3$) and 0.6% ($B = 4$).

The label rule: running time. Table 6 compares the rule with a plain enumerator *without* label dominance; this is the experiment behind the speed-ups quoted in earlier drafts, and it does not measure Algorithm A. Against Algorithm A with label dominance the result reverses (Table 7, last two columns). Added to the tiered version C5, the rule removes a further 14–32% of the extension attempts, but testing every surviving label costs more than the extensions it saves: the run becomes

slower on every label instance ($0.60\text{--}0.80\times$; $0.73\text{--}0.95\times$ with a faster implementation that takes exactly the same decisions), and on the main-grid sample it breaks even where the crowd competes (alignment 0.90) and is slower elsewhere. The reason is where it fires: 95–99% of the labels it removes already hold B orders, so each removal saves only the remaining deliveries. The rule also fires for occasional drivers (for all 30 of them in the grid sample), mostly on prefixes that can no longer reach home in time; the dead-end filter of [Proposition 4](#) removes those far more cheaply.

Table 6: Label rule against a plain enumerator without label dominance (earlier experiment; this is not a comparison with Algorithm A): median over ten timed runs [IQR], and share of label extensions saved.

Instance	$B = 3$		$B = 4$	
	Speed-up	Saved	Speed-up	Saved
$n = 12$, seed 42	1.48 [1.42, 1.57]	44.2%	2.19 [2.17, 2.22]	61.0%
$n = 10$, seed 1	1.32 [1.26, 1.36]	35.5%	1.70 [1.67, 1.70]	49.5%
$n = 15$, seed 7	2.10 [2.02, 2.19]	57.3%	3.76 [3.71, 3.79]	76.2%
$n = 12$, seed 123	1.74 [1.67, 1.81]	50.1%	2.94 [2.92, 2.94]	69.7%
$n = 10$, seed 999	1.41 [1.36, 1.53]	39.4%	1.98 [1.97, 1.98]	55.8%
Median over instances	1.48		2.19	
Pooled	1.72		2.94	

The final version of Algorithm A. [Table 7](#) reports the time to the frontier of the production code (C1), of the tiered version with the in-loop frontier (C5) and of the final version, which adds the dead-end filter (C5h, [Procedure 2](#)). The filter removes 52–76% of the extension attempts of C5, and 99% of those of occasional drivers; the frontier is unchanged on every instance, as [Proposition 4](#) guarantees, and an independent audit found no removed label with a feasible completion (12,182 labels checked). The final version is $1.9\text{--}3.9\times$ faster than the production code and $1.6\text{--}2.9\times$ faster than C5; on the main-grid sample it is $2.1\times$ (alignment 0.90) and $2.3\times$ (alignment 0.50) faster than the production code. Adding the label rule on top (C8h) is not a reliable gain: $0.81\text{--}1.03\times$ on these instances and $0.78\text{--}1.32\times$ (median $0.92\times$) over 41 instances that include the main-grid sample and $n = 20$; it is faster only where it removes at least about 40% of the remaining work, which cannot be known before the run.

Table 7: Time to the frontier (seconds, median of ten timed runs) of the production Algorithm A (C1), the tiered version with the frontier inside the loop (C5) and the final version with the dead-end filter (C5h); speed-ups are medians of paired ratios. The last two columns add the label rule to C5 (C8) and to C5h (C8h); values below 1 mean that the rule makes the run slower. Pure Python 3.13, one core of an Intel Xeon at 2.1 GHz.

Instance	B	C1	C5	C5h	C5h vs C1	C5h vs C5	C8 vs C5	C8h vs C5h
$n = 12$, seed 42	3	0.143	0.121	0.052	2.73	2.36	0.65	0.94
$n = 10$, seed 1	3	0.131	0.108	0.068	1.92	1.57	0.66	0.84
$n = 15$, seed 7	3	0.442	0.361	0.177	2.46	2.04	0.75	0.98
$n = 12$, seed 123	3	0.274	0.221	0.137	1.97	1.60	0.69	0.98
$n = 10$, seed 999	3	0.089	0.075	0.039	2.21	1.85	0.65	0.81
$n = 12$, seed 42	4	1.391	1.025	0.349	3.90	2.86	0.60	0.90
$n = 10$, seed 1	4	1.437	1.000	0.611	2.33	1.63	0.67	0.84
$n = 15$, seed 7	4	7.455	5.521	2.503	3.00	2.25	0.80	1.03
$n = 12$, seed 123	4	3.294	2.211	1.342	2.43	1.64	0.72	0.99
$n = 10$, seed 999	4	0.758	0.531	0.275	2.77	1.90	0.65	0.83

2.10 Items to verify before submission

- Read [Sol \(1994\)](#) and [Vanderbeck \(1994\)](#) directly; the statements in [Remark 9](#) are taken from [Lübbecke and Desrosiers \(2005\)](#).
- Confirm that the route deleted in [Example 2](#) belongs to the frontier of g_0 , and the exact criterion behind 556/563 in [Remark 2](#).
- State the numerical tolerances used by [Procedures 1](#) and [3](#).
- Bibliographic details of all references below.

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